



JKBOSE 2026

CLASS 11TH

STRAIGHT LINES

MATHS

LECTURE-05

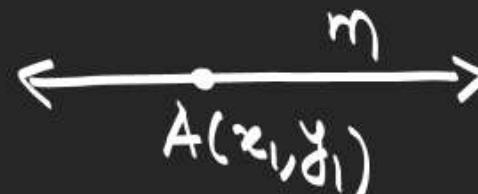
BY-MEHRAJ SIR



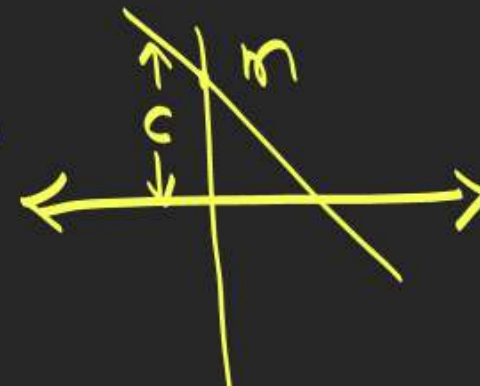
TOPICS TO BE COVERED

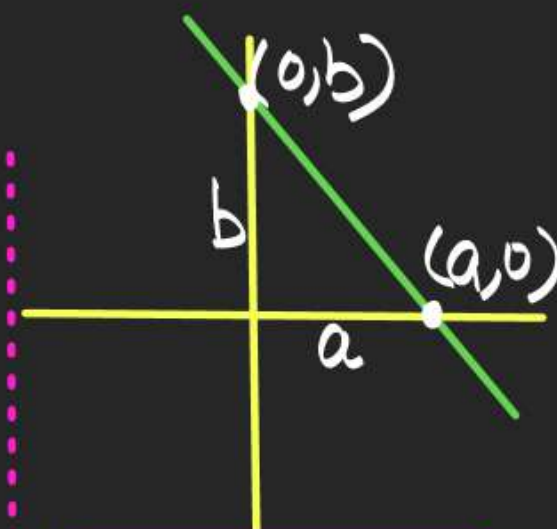
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Exercise Discussion 9.2

*  $y - y_1 = m(x - x_1)$ (Point-slope form)

*  $y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1)$ (Two point form)

*  $\rightarrow y = mx + c$ (Slope-intercept form)

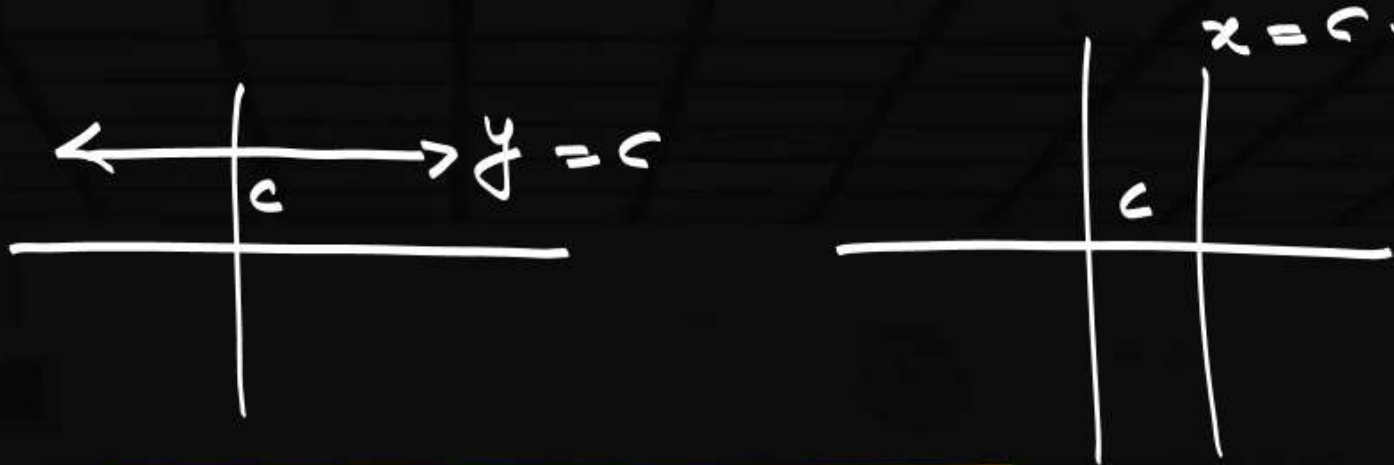


(Intercept form)
 $\frac{x}{a} + \frac{y}{b} = 1$

General form:

$$ax + by + c = 0$$

$$m = -\frac{a}{b} = -\frac{\text{Coeff. of } x}{\text{Coeff. of } y}$$

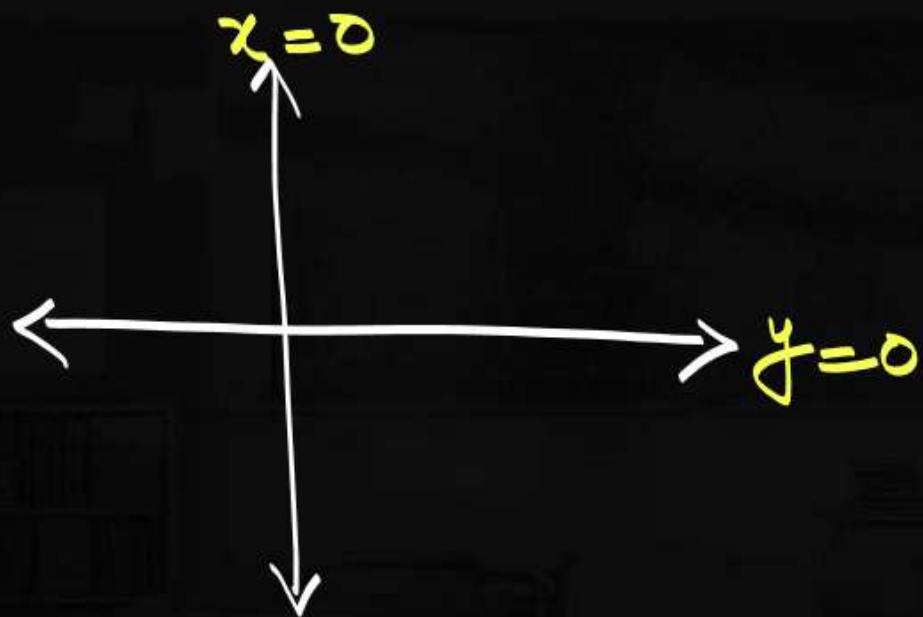


EXERCISE – 9.2

QUESTION

In Exercises 1 to 8, find the equation of the line which satisfy the given conditions:

Write the equations for the x -and y -axes.



Equation of x -axis: $y=0$

Equation of y -axis: $x=0$.

QUESTION

In Exercises 1 to 8, find the equation of the line which satisfy the given conditions:

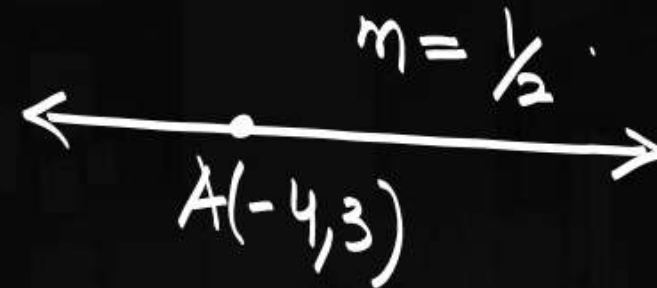
Passing through the point $(-4, 3)$ with slope $1/2$.

(Sol) Required equation is:

$$y - 3 = \frac{1}{2}(x + 4)$$

$$\Rightarrow 2y - 6 = x + 4$$

$$\Rightarrow x - 2y + 10 = 0 \quad (\text{Hence the result})$$



QUESTION

In Exercises 1 to 8, find the equation of the line which satisfy the given conditions:

Passing through $(0, 0)$ with slope m .

$$\begin{aligned}y - y_1 &= m(x - x_1) \\ \Rightarrow y - 0 &= m(x - 0) \\ \Rightarrow y &= mx.\end{aligned}$$

QUESTION

In Exercises 1 to 8, find the equation of the line which satisfy the given conditions:

Passing through $(2, 2\sqrt{3})$ and inclined with the x-axis at an angle of 75°

(sol) Since $\theta = 75^\circ$ is the angle of inclination, so

Slope of the line is given by;

$$m = \tan \theta = \tan 75^\circ$$

$$\Rightarrow m = \tan(30^\circ + 45^\circ)$$

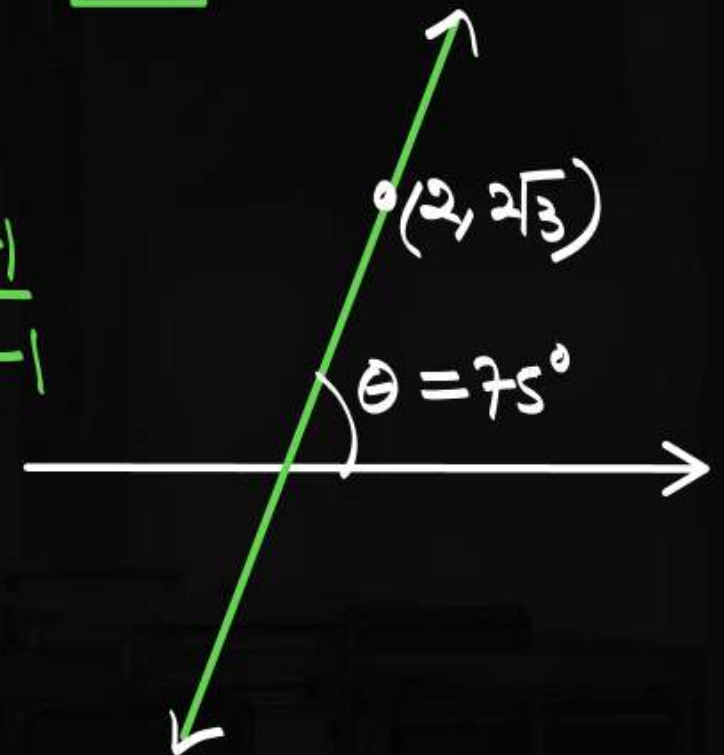
$$\Rightarrow m = \frac{\tan 30^\circ + \tan 45^\circ}{1 - \tan 30^\circ \tan 45^\circ}$$

$$\Rightarrow m = \frac{\frac{1}{\sqrt{3}} + 1}{1 - \frac{1}{\sqrt{3}}}$$

$$\Rightarrow m = \frac{\left(\frac{1 + \sqrt{3}}{\sqrt{3}}\right)}{\left(\frac{\sqrt{3} - 1}{\sqrt{3}}\right)} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$$

$$\Rightarrow m = \frac{(\sqrt{3} + 1)(\sqrt{3} + 1)}{(\sqrt{3} - 1)(\sqrt{3} + 1)}$$

$$\Rightarrow m = \frac{(\sqrt{3} + 1)^2}{\sqrt{3}^2 - 1^2} = \frac{\sqrt{3}^2 + 1^2 + 2\sqrt{3}}{2}$$



$$\Rightarrow m = \frac{4+2\sqrt{3}}{2}$$

$$\Rightarrow m = 2 + \sqrt{3}$$

Therefore the required equation is:

$$y - 2\sqrt{3} = (2 + \sqrt{3})(x - 2)$$

$$\Rightarrow y - 2\sqrt{3} = (2 + \sqrt{3})x - 4 - 2\sqrt{3}$$

$$\Rightarrow (2 + \sqrt{3})x - y - 4 = 0$$

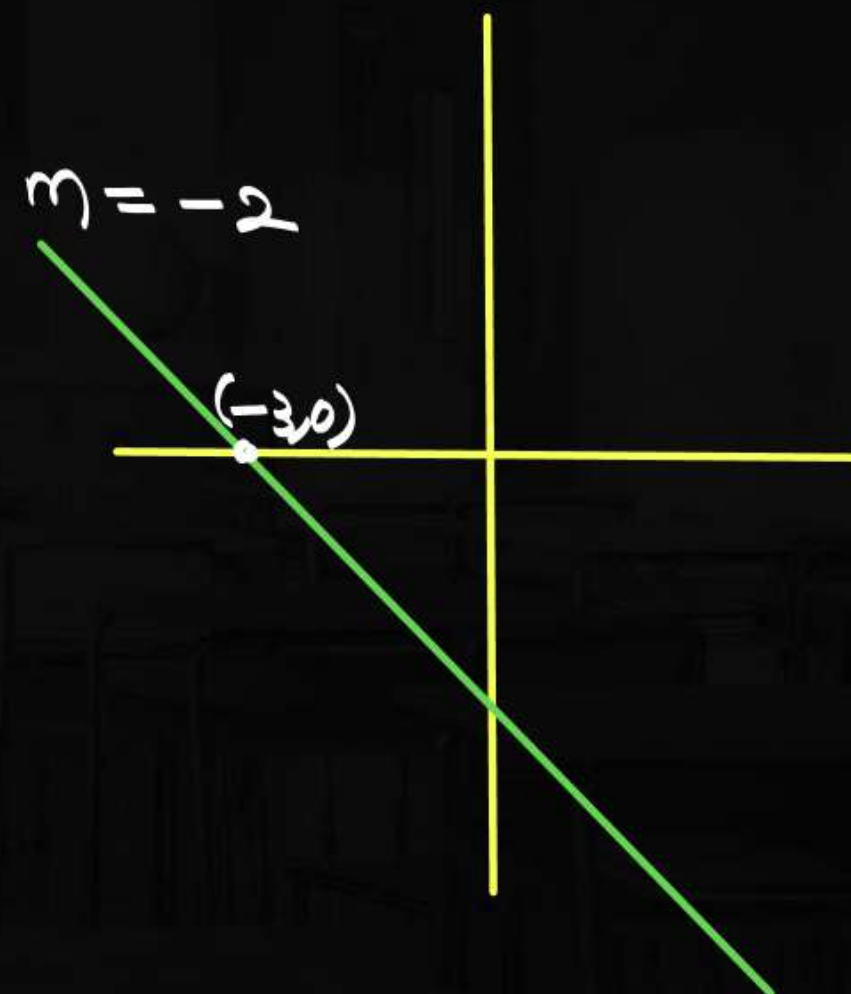
Hence the result.

QUESTION

In Exercises 1 to 8, find the equation of the line which satisfy the given conditions:

Intersecting the **x-axis** at a distance of **3 units** to the **left of origin** with **slope -2**.

$$\begin{aligned} y - 0 &= -2(x + 3) \\ \Rightarrow y &= -2x - 6. \\ \Rightarrow 2x + y + 6 &= 0. \end{aligned}$$

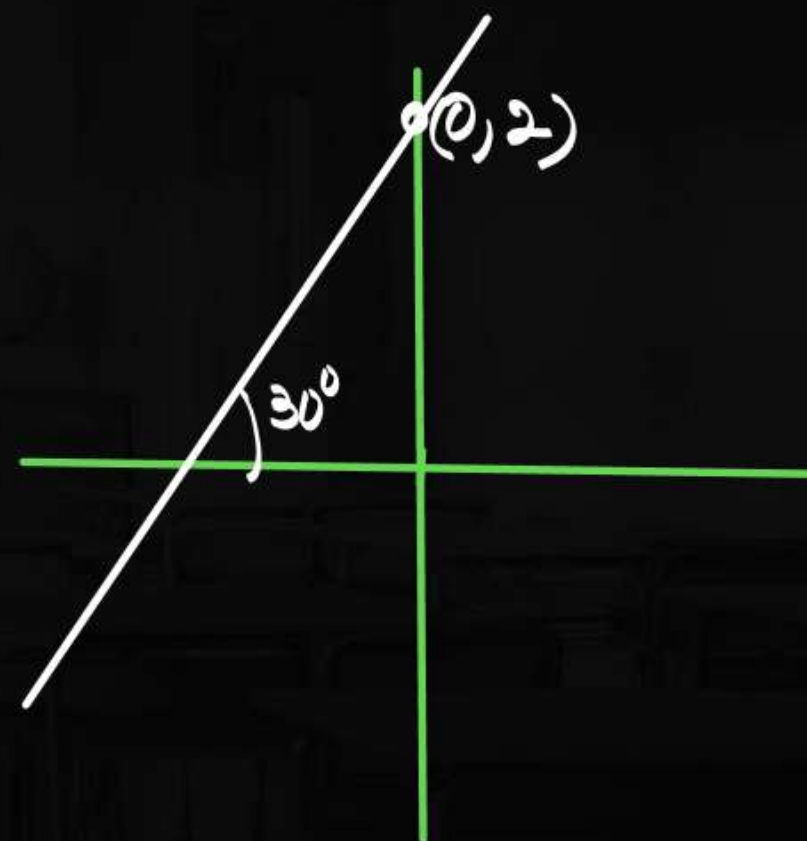


QUESTION

In Exercises 1 to 8, find the equation of the line which satisfy the given conditions:

Intersecting the y -axis at a distance of 2 units above the origin and making an angle of 30° with positive direction of the x -axis.

$$\begin{aligned}
 y &= (\tan 30^\circ)x + 2 \\
 \Rightarrow y &= \frac{x}{\sqrt{3}} + 2 \\
 \Rightarrow \sqrt{3}y &= x + 2 \\
 \Rightarrow x - \sqrt{3}y + 2 &= 0
 \end{aligned}$$



QUESTION

In Exercises 1 to 8, find the equation of the line which satisfy the given conditions:

Passing through the points $(-1, 1)$ and $(2, -4)$.

(Sol) The Required equation of the line is:

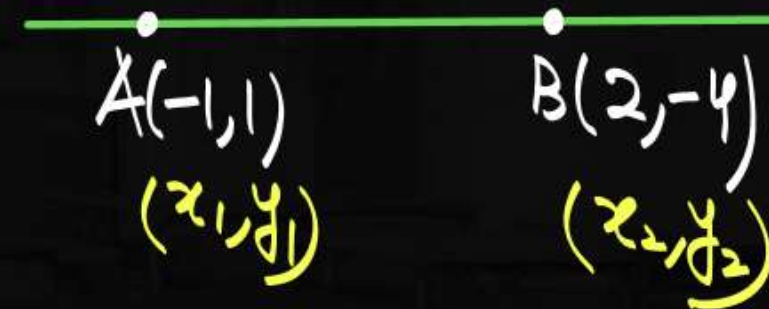
$$y - 1 = \left(\frac{-4 - 1}{2 + 1} \right) (x + 1)$$

$$\Rightarrow y - 1 = -\frac{5}{3}(x + 1)$$

$$\Rightarrow 3y - 3 = -5x - 5$$

$$\Rightarrow 5x + 3y + 2 = 0$$

Hence the result.



QUESTION

In Exercises 1 to 8, find the equation of the line which satisfy the given conditions:

The vertices of ΔPQR are $P(2, 1)$, $Q(-2, 3)$ and $R(4, 5)$. Find equation of the median through the vertex R .

(Sol) Suppose that M is the Mid-point of the side PQ , then RM is the Median.

Here By Mid-point formula;

$$M(x, y) = M\left(\frac{-2+2}{2}, \frac{1+3}{2}\right)$$

$$\Rightarrow M(x, y) = M(0, 2)$$

so the required equation is:

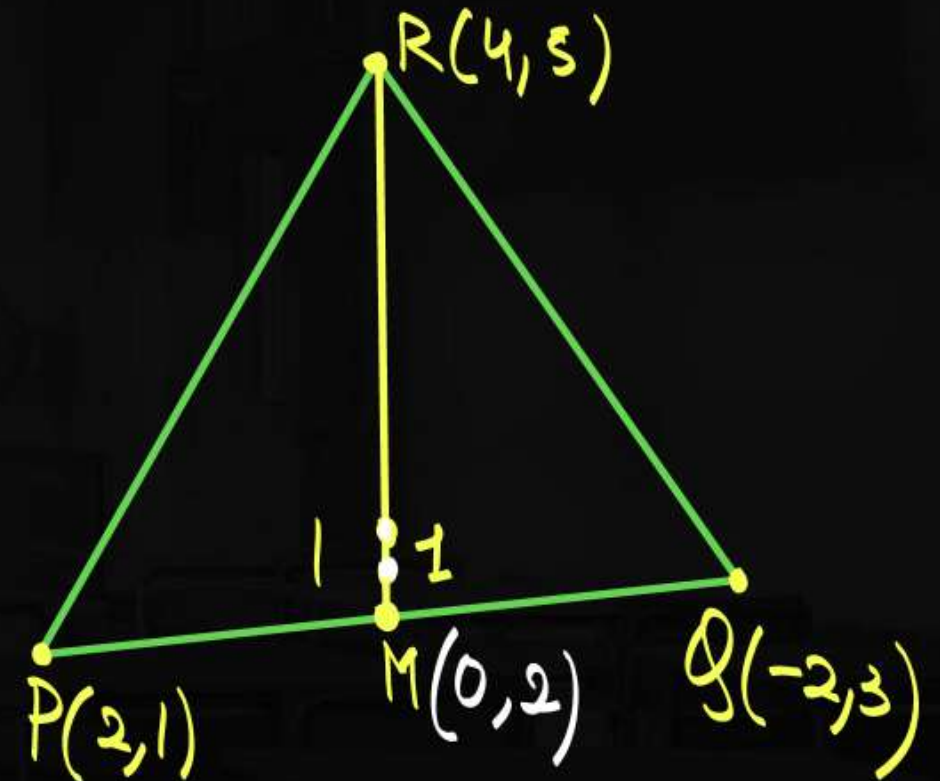
$$y - 2 = \frac{5-2}{4-0}(x - 0)$$

$$\Rightarrow y - 2 = \frac{3}{4}x$$

$$\Rightarrow 4y - 8 = 3x$$

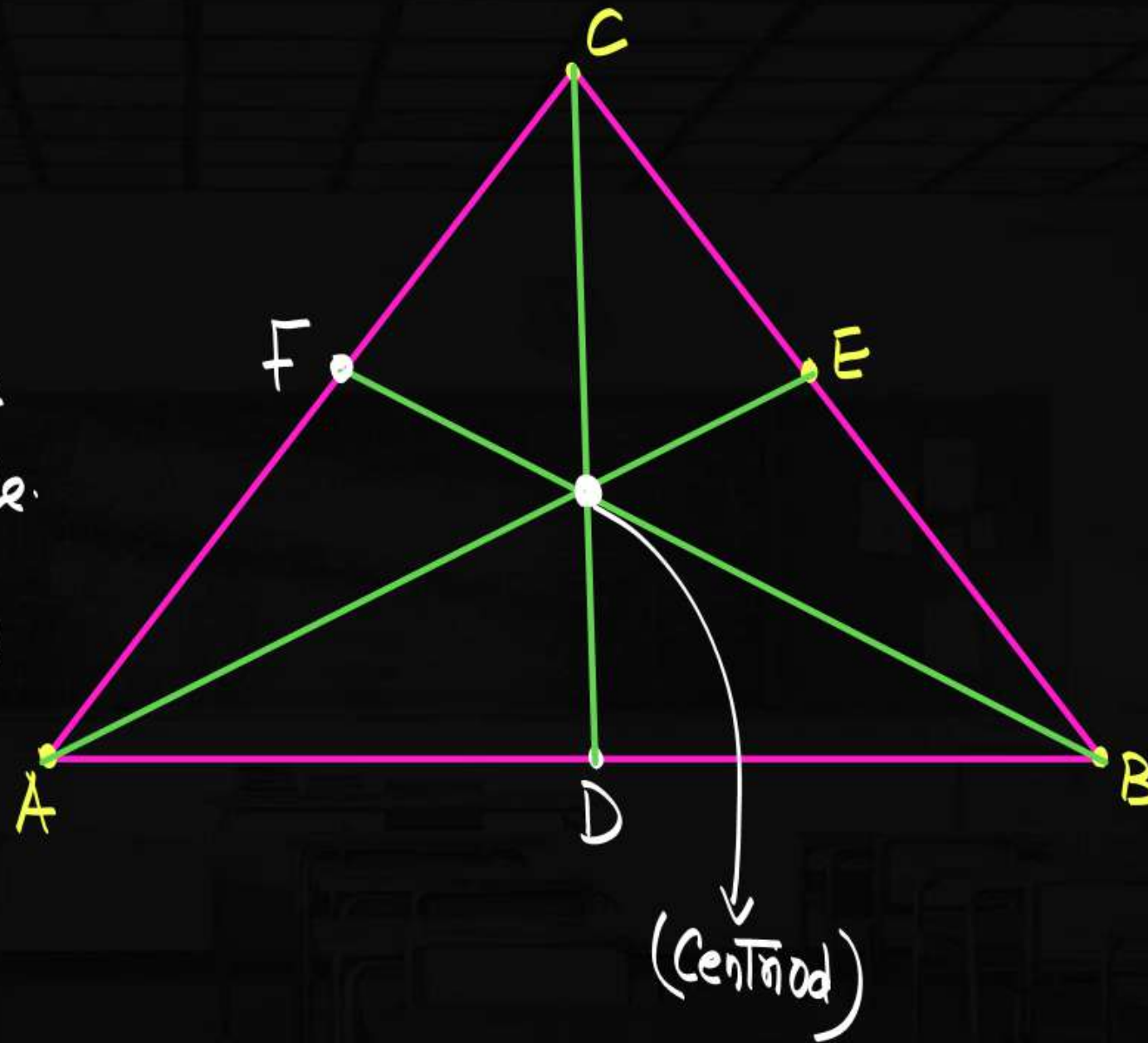
$$\Rightarrow \boxed{3x - 4y + 8 = 0}$$

Hence the result.



AE, BF, CD are the medians of the Triangle.

* Point of Concurrence of the medians is known as the Centroid.



Line segment joining the vertex with the mid-point of the side opposite to it is known as a median.

QUESTION

Find the equation of the line passing through $(-3, 5)$ and perpendicular to the line through the points $(2, 5)$ and $(-3, 6)$.

(Sol) We know that the product of the slopes of the perpendicular lines is -1 .

$$\text{So, } m_{AB} \times m = -1.$$

$$\Rightarrow \left(\frac{6-5}{-3-2} \right) \times m = -1.$$

$$\Rightarrow \frac{1}{-5} \times m = -1$$

$$\Rightarrow m = 5$$

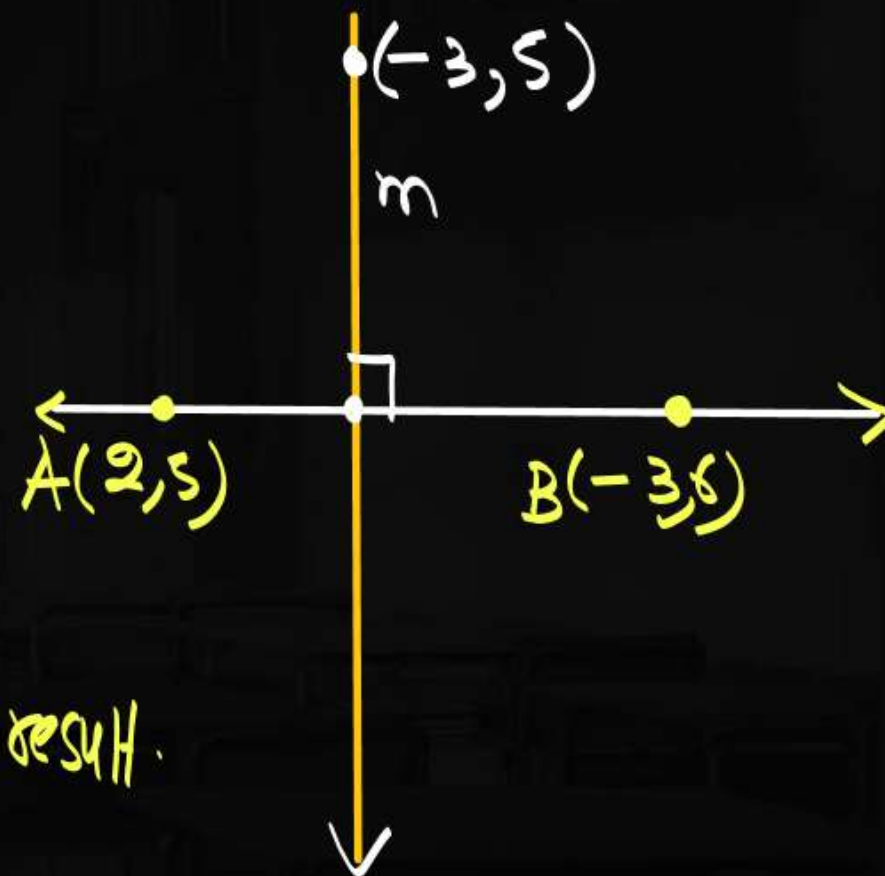
So, the required equation is:

$$y - 5 = 5(x + 3)$$

$$\Rightarrow y - 5 = 5x + 15.$$

$$\Rightarrow 5x - y + 20 = 0$$

Hence the result.



QUESTION

A line perpendicular to the line segment joining the points $(1, 0)$ and $(2, 3)$ divides it in the ratio $1 : n$. Find the equation of the line.

(Sol) We know that the product of the slopes of perpendicular lines is -1 . So

$$m_{AB} \times m = -1$$

$$\Rightarrow \left(\frac{3-0}{2-1} \right) \times m = -1$$

$$\Rightarrow 3m = -1$$

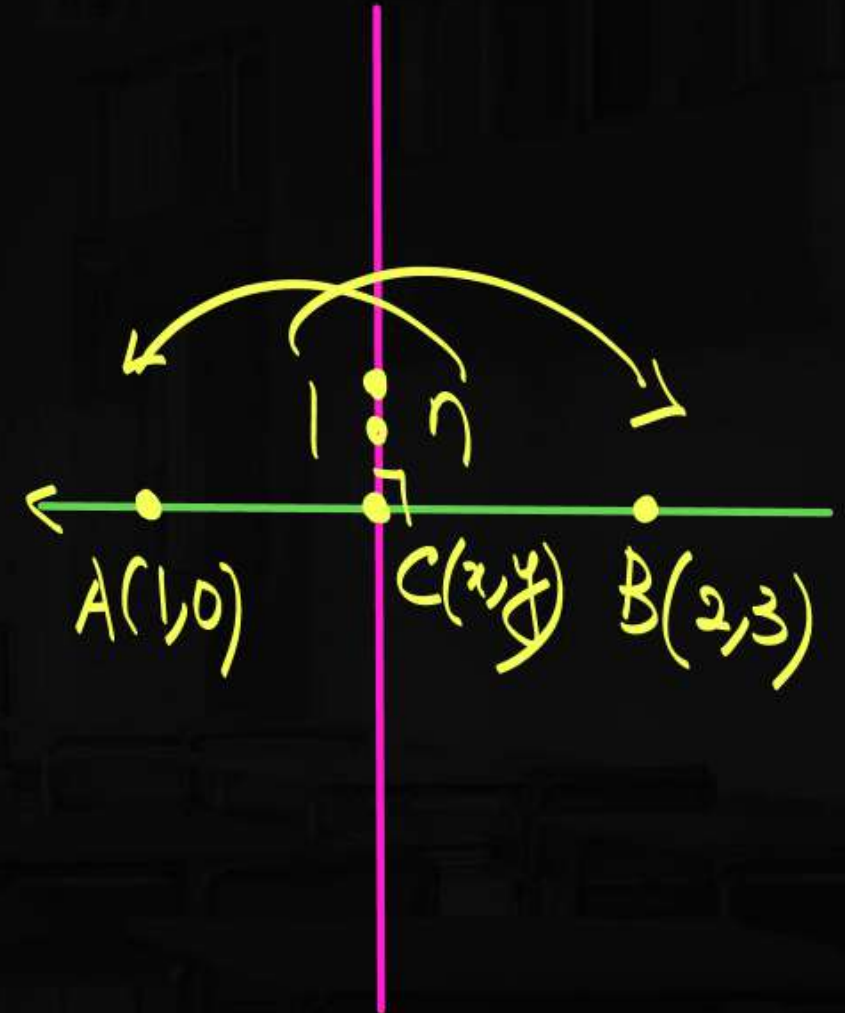
$$\Rightarrow m = -\frac{1}{3}$$

Here $C(x, y)$ divides AB in the ratio $1:n$,
So by section formula,

$$C(x, y) = \left(\frac{n+2}{n+1}, \frac{3}{n+1} \right)$$

So, the required equation is:

$$y - \frac{3}{n+1} = -\frac{1}{3} \left(x - \frac{n+2}{n+1} \right)$$



$$\Rightarrow 3y - \frac{9}{n+1} = -x + \frac{n+2}{n+1}$$

$$\Rightarrow x + 3y - \frac{9}{n+1} - \frac{n+2}{n+1} = 0.$$

$$\Rightarrow x + 3y - \left(\frac{9+n+2}{n+1} \right) = 0$$

$$\Rightarrow x + 3y - \left(\frac{11+n}{n+1} \right) = 0.$$

QUESTION

Find the equation of a line that cuts off equal intercepts on the coordinate axes and passes through the point $(2, 3)$.

(Sol) let the required equation be

$$\frac{x}{a} + \frac{y}{a} = 1$$

$$\Rightarrow x + y = a \quad \text{--- ①}$$

Since $(2, 3)$ lies on ①, so $2 + 3 = a$.

$$\Rightarrow a = 5.$$

Therefore the req. eqn is:

$$x + y = 5.$$

QUESTION

Find equation of the line passing through the point $(2, 2)$ and cutting off intercepts on the axes whose sum is 9.

(Sol) let the required equation be:

$$\frac{x}{a} + \frac{y}{b} = 1 \quad \text{--- (1)}$$

Here, it's given that $a+b=9$.

$$\Rightarrow b = 9 - a \quad \text{--- (2)}$$

Also, since (1) passes through $(2, 2)$. So,

$$\frac{2}{a} + \frac{2}{b} = 1$$

$$\Rightarrow 2b + 2a = ab \quad \text{--- (3)}$$

Using (2) in (3), we get:

$$2(9-a) + 2a = a(9-a)$$

$$\Rightarrow 18 - \cancel{2a} + \cancel{2a} = 9a - a^2$$

$$\Rightarrow a^2 - 9a + 18 = 0$$

$$\Rightarrow (a-6)(a-3) = 0$$

$$\Rightarrow a = 6, 3$$

$$\text{So, } b = 3, 6$$

So from (1), the req. equations are:

$$\frac{x}{6} + \frac{y}{3} = 1$$

$$\frac{x}{3} + \frac{y}{6} = 1$$

$$\Rightarrow 3x + 6y = 18$$

$$\Rightarrow 6x + 3y = 18$$

$$\Rightarrow x + 2y = 6$$

$$2x + y = 6$$

QUESTION

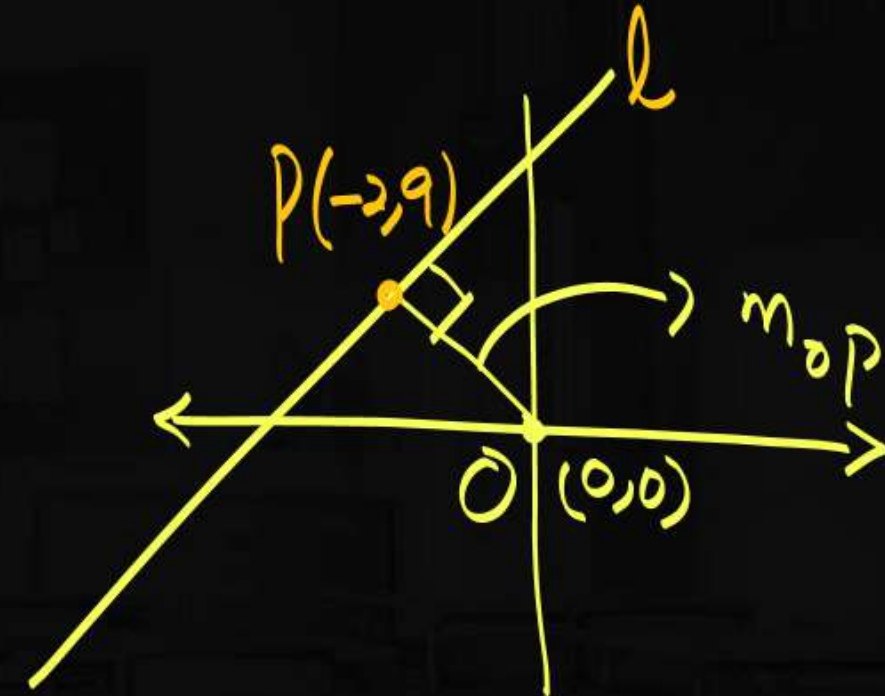
Find equation of the line through the point $(0, 2)$ making an angle $2\pi/3$ with the positive x -axis. Also, find the equation of line parallel to it and crossing the y -axis at a distance of 2 units below the origin.

(Homework)

QUESTION

The perpendicular from the origin to a line meets it at the point $(-2, 9)$, find the equation of the line.

$$m_{op} \times m = -1.$$



QUESTION

The length L (in centimetre) of a copper rod is a linear function of its Celsius temperature C . In an experiment, if $L = 124.942$ when $C = 20$ and $L = 125.134$ when $C = 110$ express L in terms of C .

(Home work)

$$y = mx + c$$

$$L = aC + b$$

$$124.942 = 20a + b \quad \text{--- (1)}$$

H/W

$$125.134 = 110a + b \quad \text{--- (2)}$$

QUESTION

$$(14, 980) \\ (16, 1220)$$

$$(P, D) = (14, 980), (16, 1220)$$

The owner of a milk store finds that, he can sell 980 litres of milk each week at Rs 14/litre and 1220 litres of milk each week at Rs 16/litre. Assuming a linear relationship between selling price and demand, how many litres could he sell weekly at Rs 17/litre?

(Sol) let P & D be respectively the selling price & Demand. Then,

$$D = aP + b \quad \text{--- (1)}$$

Since $(14, 980)$ & $(16, 1220)$ lies on (1), so;

$$\begin{aligned} 980 &= 14a + b \\ 1220 &= 16a + b \end{aligned} \quad \rightarrow \text{Solve for finding } a \& b.$$

$$D = a \times 17 + b$$

QUESTION

$P(a, b)$ is the mid-point of a line segment between axes. Show that equation of the line is $\frac{x}{a} + \frac{y}{b} = 2$.

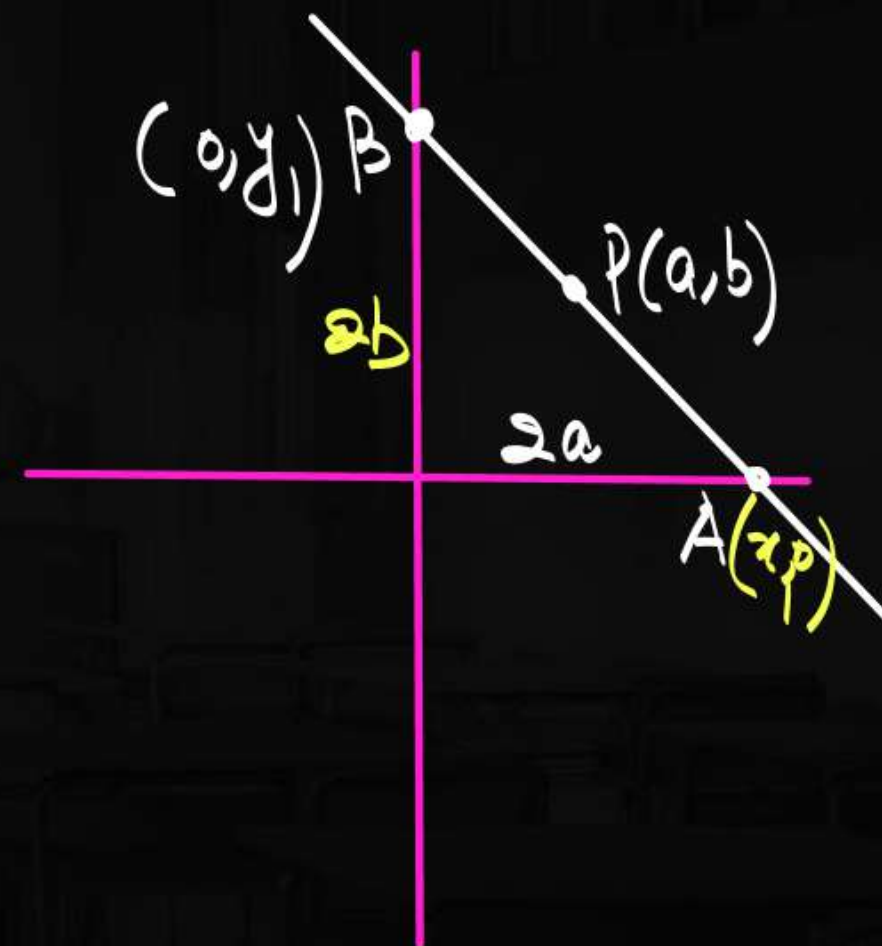
Proof: Since $P(a, b)$ is the mid-point of AB

$$\text{So, } a = \frac{x_1 + 0}{2}, \quad b = \frac{y_1 + 0}{2}$$

$$\Rightarrow x_1 = 2a, \quad y_1 = 2b$$

Therefore the required equation is;

$$\frac{x}{2a} + \frac{y}{2b} = 1 \Rightarrow \boxed{\frac{x}{a} + \frac{y}{b} = 2} \quad \text{Hence proved.}$$



QUESTION

Point $R(h, k)$ divides a line segment between the axes in the ratio $1 : 2$.

Find equation of the line.

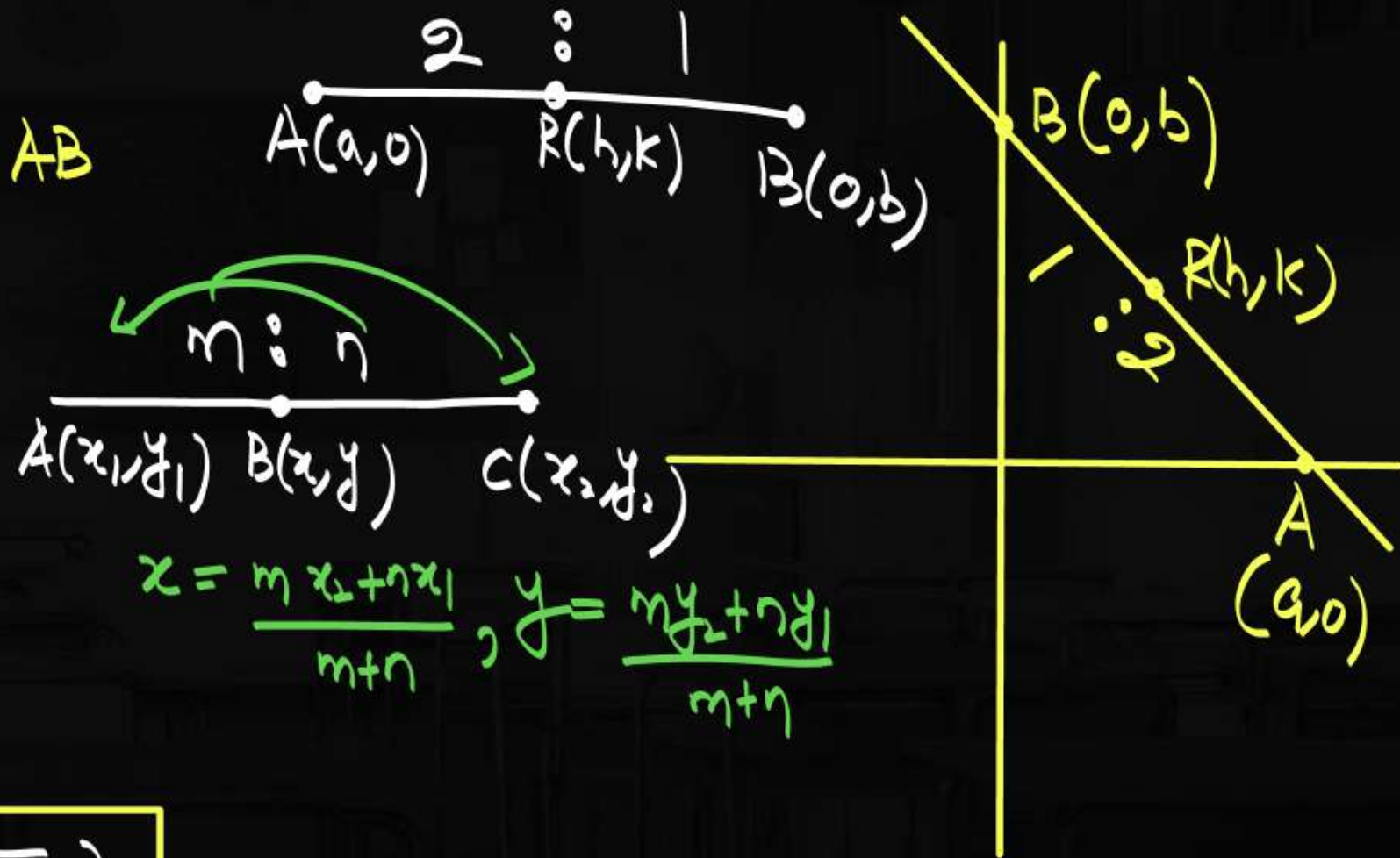
(Sol) Since $R(h, k)$ divides the line segment AB in the ratio of $1:2$. So,

$$h = \frac{a+0}{1+2}, \quad k = \frac{0+2b}{1+2}$$

$$\Rightarrow a = 3h, \quad b = \frac{3k}{2}$$

So, the required equation is:

$$\frac{x}{3h} + \frac{y}{\left(\frac{3k}{2}\right)} = 1 \Rightarrow \boxed{\frac{x}{h} + \frac{2y}{k} = 3}$$



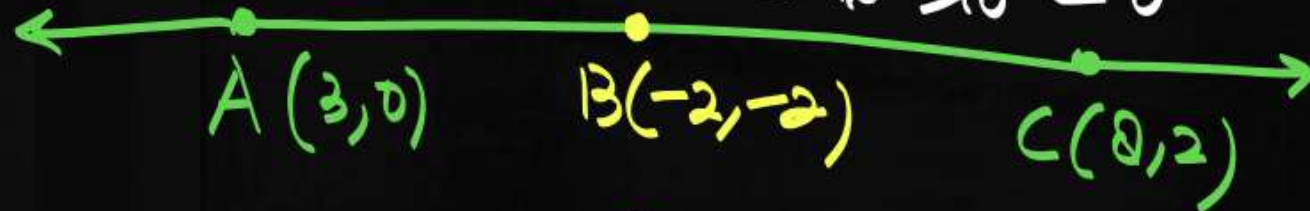
QUESTION

By using the concept of equation of a line, prove that the three points $(3, 0)$, $(-2, -2)$ and $(8, 2)$ are collinear.

$$\begin{aligned} 2x - 2 - 5x - 2 - 6 \\ = -4 + 10 - 6 \\ = 10 - 10 = 0 \end{aligned}$$

(Homework)

$$A(3,0), B(-2,-2) \\ C(8,2)$$



$$y - 0 = \frac{2 - 0}{8 - 3}(x - 3)$$

$$\Rightarrow y = \frac{2}{5}(x - 3)$$

$$\Rightarrow 5y = 2x - 6$$

$$\Rightarrow 2x - 5y - 6 = 0 \quad \text{①}$$

Here clearly $B(-2,-2)$ satisfies ①
So $A(3,0)$, $B(-2,-2)$ & $C(8,2)$
are collinear points.
#

Thank
*Co-ordinate
Geometry* **You**